

OR

```
mysql> SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player WHERE LName='Tendulkar');
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT MatchID FROM Batting,Player WHERE PID=PlayerID AND Lname='Tendulkar';
+-----+
| MatchID |
+-----+
| 2689 |
+-----+
1 row in set (0.0007 sec)
```

Q2. Find match information of all matches in which Dohni has batted.

```
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch,Batting,Player;
```

```
+-----+-----+-----+-----+
| India      | England | Delhi  | 2006-03-28 |
| Pakistan   | India   | Rawalpindi | 2006-02-11 |
| Pakistan   | India   | Rawalpindi | 2006-02-11 |
| Pakistan   | India   | Rawalpindi | 2006-02-11 |
| Pakistan   | India   | Rawalpindi | 2006-02-11 |
| Pakistan   | India   | Rawalpindi | 2006-02-11 |
| Pakistan   | India   | Rawalpindi | 2006-02-11 |
| Pakistan   | India   | Rawalpindi | 2006-02-11 |
| Pakistan   | India   | Rawalpindi | 2006-02-11 |
| Pakistan   | India   | Rawalpindi | 2006-02-11 |
| Pakistan   | India   | Rawalpindi | 2006-02-11 |
| Pakistan   | India   | Rawalpindi | 2006-02-11 |
| Pakistan   | India   | Rawalpindi | 2006-02-11 |
| Pakistan   | India   | Rawalpindi | 2006-02-11 |
| Pakistan   | India   | Rawalpindi | 2006-02-11 |
| Pakistan   | India   | Rawalpindi | 2006-02-11 |
| Pakistan   | India   | Rawalpindi | 2006-02-11 |
| Pakistan   | India   | Peshwar  | 2006-02-06 |
| Pakistan   | India   | Peshwar  | 2006-02-06 |
| Pakistan   | India   | Peshwar  | 2006-02-06 |
| Pakistan   | India   | Peshwar  | 2006-02-06 |
| Pakistan   | India   | Peshwar  | 2006-02-06 |
| Pakistan   | India   | Peshwar  | 2006-02-06 |
| Pakistan   | India   | Peshwar  | 2006-02-06 |
| Pakistan   | India   | Peshwar  | 2006-02-06 |
| Pakistan   | India   | Peshwar  | 2006-02-06 |
| Pakistan   | India   | Peshwar  | 2006-02-06 |
| Pakistan   | India   | Peshwar  | 2006-02-06 |
| Pakistan   | India   | Peshwar  | 2006-02-06 |
| Pakistan   | India   | Peshwar  | 2006-02-06 |
| Pakistan   | India   | Peshwar  | 2006-02-06 |
| Pakistan   | India   | Peshwar  | 2006-02-06 |
| Pakistan   | India   | Peshwar  | 2006-02-06 |
| Pakistan   | India   | Peshwar  | 2006-02-06 |
+-----+-----+-----+-----+
9280 rows in set (0.0131 sec)
```

```
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch WHERE MatchID IN (SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player WHERE LName ='Dhoni'));
```

OR

Using Join

```
mysql> SELECT Team1,Team2,Ground,Date FROM ODIMatch,Batting,Player WHERE ODIMatch.MatchID=Batting.MatchID AND PlayerID=PID AND LName='Dhoni';
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT Team1,Team2,Ground,Date FROM ODIMatch WHERE MatchID IN (SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player WHERE LName='Dhoni'));
```

Team1	Team2	Ground	Date
Sri Lanka	India	Colombo	2008-08-24
Australia	India	Brisbane	2008-03-04

2 rows in set (0.0014 sec)

Q3. Find match Ids of all matches in which Tendulkar batted.

```
mysql> SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player WHERE Lname='Tendulkar'); OR
```

Using Join

```
mysql> SELECT MatchID FROM Batting,Player WHERE PID=PlayerID AND LName='Tendulkar';
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM Player WHERE Lname='Tendulkar');
```

MatchID
2689

1 row in set (0.0008 sec)

Q4. Find Ids of all players that have both bowled and batted in the ODI match 2689.

```
mysql> SELECT PID FROM Batting WHERE MatchID=2689 AND PID IN (SELECT PID FROM Bowling WHERE MatchID=2689);
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT PID FROM Batting WHERE MatchID=2689 AND PID IN (SELECT PID FROM Bowling WHERE MatchID=2689);
```

PID
99001
24001
23001
98002

4 rows in set (0.0139 sec)

Q5. Find Ids of all players that have either bowled or batted (or did both) in ODI match 2689.

```
mysql> SELECT PID FROM Batting WHERE MatchID=2689 OR PID IN (SELECT PID FROM Bowling WHERE MatchID=2689);
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT PID FROM Batting WHERE MatchID=2689 OR PID IN (SELECT PID FROM Bowling WHERE MatchID=2689);
```

PID
23001
89001
23001
25001
99002
95001
24001
99001
27001
98002
98002

11 rows in set (0.0037 sec)

Using Union

mysql> SELECT PID FROM Batting WHERE MatchID=2689 UNION (SELECT PID FROM Bowling WHERE MatchID=2689);

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT PID FROM Batting WHERE MatchID=2689 UNION (SELECT PID FROM Bowling WHERE MatchID=2689);
+----+
| PID |
+----+
| 89001 |
| 23001 |
| 25001 |
| 99002 |
| 95001 |
| 24001 |
| 99001 |
| 27001 |
| 98002 |
+----+
9 rows in set (0.0008 sec)
```

Q6. Find Ids.

mysql> SELECT PID FROM Batting WHERE MatchID=2689 AND PID NOT IN (SELECT PID FROM Bowling WHERE MatchID=2689);

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT PID FROM Batting WHERE MatchID=2689 AND PID NOT IN (SELECT PID FROM Bowling WHERE MatchID=2689);
+----+
| PID |
+----+
| 89001 |
| 25001 |
| 99002 |
| 95001 |
| 27001 |
+----+
5 rows in set (0.0007 sec)
```

Q7. Find match Ids where Tendulkar has played.

mysql> SELECT MatchID FROM Batting,(SELECT PlayerID FROM Player WHERE LName='Tendulkar') ST WHERE PID=ST.PlayerID;

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT MatchID FROM Batting,(SELECT PlayerID FROM Player WHERE LName='Tendulkar') ST WHERE PID=ST.PlayerID;
+----+
| MatchID |
+----+
| 2689 |
+----+
1 row in set (0.0027 sec)
```

Q8. Find match Ids of all those matches in which player 27001 bats and makes more runs than he made at every match he played at Colombo.

mysql> SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns > ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo'));

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns > ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo'));
+----+
| MatchID |
+----+
| 2689 |
+----+
1 row in set (0.0028 sec)
```

Q9(a). Find Ids and scores of players who scored less than 75 but more than 50 in Colombo.

mysql> SELECT PID,NRuns FROM Batting WHERE NRuns BETWEEN 51 AND 74 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo');

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT MatchID FROM Batting WHERE PID=27001 AND NRuns < ALL (SELECT NRuns FROM Batting WHERE PID=27001 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Brisbane'));
```

MatchID
2755

Q9(b). Find Ids and scores of players who scored greater than 49 runs in Colombo.

mysql> SELECT PID,NRuns FROM Batting WHERE NRuns > 49 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo');

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT PID,NRuns FROM Batting WHERE NRuns BETWEEN 51 AND 74 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo');
```

PID	NRuns
25001	71
91001	60

2 rows in set (0.0015 sec)

Q10. Find the player Ids of those players whose date of first match (FTest) is not given in the database.

mysql> SELECT PlayerID FROM Player WHERE FTest IS NULL;

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT PID,NRuns FROM Batting WHERE NRuns > 49 AND MatchID IN (SELECT MatchID FROM ODIMatch WHERE Ground='Colombo');
```

PID	NRuns
25001	71
91001	60

2 rows in set (0.0004 sec)

Q11. Find names of the player who batted in match 2689.

mysql> SELECT FName,LName FROM Player WHERE EXISTS (SELECT * FROM Batting WHERE PlayerID=PID AND MatchID=2689);

OR

mysql> SELECT FName,LName FROM Player,Batting WHERE PlayerID = PID AND MatchID=2689;

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT PlayerID FROM Player WHERE FTest IS NULL;
```

PlayerID
27001

1 row in set (0.0005 sec)

Q12. Find player Ids of player who have scored more than 30 in every ODI match that they have played.

mysql> SELECT PID AS PlayerID FROM Batting b1 WHERE NOT EXISTS (SELECT * FROM Batting b2 WHERE b1.PID=b2.PID AND NRuns < 31);

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT FName,LName FROM Player WHERE EXISTS ( SELECT * FROM Batting WHERE PlayerID=PID AND MatchID=2689);
```

FName	LName
Sachin	Tendulkar
Yuvraj	Singh
MS	Dhoni
Adam	Gilchrist
Ricky	Ponting
Andrew	Symonds
Breet	Lee
Praveen	Kumar
Harbhajan	Singh

```
9 rows in set (0.0005 sec)
```

Q13. Find the names of players that have batted in all the ODI matches in Melbourne that are in the database.

```
SELECT FName, LName
FROM Player
WHERE NOT EXISTS
(SELECT *
FROM ODIMatch
WHERE Ground = 'Melbourne'
AND NOT EXISTS
(SELECT *
FROM Batting
WHERE PID=PlayerID
AND (Batting.MatchId=ODIMatch.MatchID));
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT FName, LName FROM Player WHERE NOT EXISTS (SELECT * FROM ODIMatch WHERE Ground = 'Melbourne' AND NOT EXISTS (SELECT * FROM Batting WHERE Batting.PID = Player.PlayerID AND Batting.MatchID = ODIMatch.MatchID));
```

Empty set (0.0007 sec)

Q14. Find the number of players that bowled in ODI match 2689.

```
mysql> SELECT COUNT(*) AS NBowlers FROM Bowling WHERE MatchID=2689;
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT AVG(NRuns) AS AvgRuns_2689 FROM Batting WHERE MatchID=2689;
```

AvgRuns_2689
25.2222

```
1 row in set (0.0031 sec)
```

Q15. Find the average batting score of all the players that batted in the ODI Match 2689.

```
mysql> SELECT AVG(NRuns) AS AvgRuns_2689 FROM Batting WHERE MatchID=2689;
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT AVG(NRuns) AS AvgRuns_2689 FROM Batting WHERE MatchID=2689;
```

AvgRuns_2689
25.2222

```
1 row in set (0.0031 sec)
```

Q16. Find the youngest player in the database.

```
mysql> SELECT FName AS Youngest_Player FROM Player
WHERE YBorn=(SELECT MAX(YBorn) FROM Player);
```

```

MySQL localhost:3306 ssl mahek01 SQL > SELECT FName AS Youngest_Player FROM Player WHERE YBorn=(SELECT MAX(YBorn) FROM Player);
+-----+
| Youngest_Player |
+-----+
| Ishant          |
+-----+
1 row in set (0.0031 sec)

```

Q17. Find the number of players in the database from each country.

mysql> SELECT Country, COUNT(*) AS NPlayers FROM Player GROUP BY Country;

```

MySQL localhost:3306 ssl mahek01 SQL > SELECT Country, COUNT(*) AS NPlayers FROM Player GROUP BY Country;
+-----+-----+
| Country | NPlayers |
+-----+-----+
| India   | 10       |
| Australia | 5       |
| West Indies | 1       |
| Sri Lanka | 3       |
| Pakistan | 2       |
| Zimbabwe | 1       |
| South Africa | 4       |
| New Zealand | 2       |
| England | 1       |
+-----+-----+
9 rows in set (0.0007 sec)

```

Q18. Find batting average of each player in the database. Present the PID of each Player.

mysql> SELECT PID,AVG(NRuns) AS AVGR FROM Batting GROUP BY PID;

```

MySQL localhost:3306 ssl mahek01 SQL > SELECT PID,AVG(NRuns) AS AVGR FROM Batting GROUP BY PID;
+-----+-----+
| PID | AVGR |
+-----+-----+
| 23001 | 19.0000 |
| 24001 | 42.0000 |
| 25001 | 53.5000 |
| 27001 | 4.5000 |
| 89001 | 91.0000 |
| 91001 | 60.0000 |
| 92002 | 1.0000 |
| 94002 | 17.0000 |
| 95001 | 1.0000 |
| 98002 | 2.5000 |
| 99001 | 7.0000 |
| 99002 | 2.0000 |
+-----+-----+
12 rows in set (0.0006 sec)

```

Q19. Find batting average of each player from India in the database. Present first name of each player.

mysql> SELECT PID,AVG(NRuns) AS AVGR FROM Batting GROUP BY PID;

```

MySQL localhost:3306 ssl mahek01 SQL > SELECT P.FName, SUM(B.NRuns) / COUNT(B.MatchID) AS Batting_Average FROM Player P JOIN Batting B ON P.PlayerID = B.PID WHERE P.Country = 'India' GROUP BY P.PlayerID, P.FName;
+-----+-----+
| FName | Batting_Average |
+-----+-----+
| Yuvraj | 19.0000 |
| MS     | 53.5000 |
| Praveen | 4.5000 |
| Sachin | 91.0000 |
| Harbhajan | 2.5000 |
+-----+-----+
5 rows in set (0.0029 sec)

```

Q20. Find the average score for each player when playing against Australia.

```
mysql> SELECT P.FName, SUM(B.NRuns) / COUNT(B.MatchID) AS Batting_Average
FROM Player P JOIN Batting B ON P.PlayerID = B.PID WHERE P.Country = 'India'
GROUP BY P.PlayerID, P.FName;
```

```
5 rows in set (0.0029 sec)
MySQL localhost:3306 ssl mahek01 SQL > SELECT P.FName, AVG(B.NRuns) AS Average_Score FROM Player P JOIN Batting B ON P.PlayerID = B.PID JOIN ODI
Match M ON B.MatchID = M.MatchID WHERE M.Team1 = 'Australia' OR M.Team2 = 'Australia' GROUP BY P.PlayerID, P.FName;
+-----+-----+
| FName | Average_Score |
+-----+-----+
| Sachin | 91.0000 |
| Yuvraj | 38.0000 |
| MS | 36.0000 |
| Adam | 2.0000 |
| Ricky | 1.0000 |
| Andrew | 42.0000 |
| Breet | 7.0000 |
| Praveen | 7.0000 |
| Harbhajan | 3.0000 |
+-----+-----+
9 rows in set (0.0080 sec)
```

Q21. Find the ID of players that had a higher average score for all players when they played against Sri Lanka.

```
SELECT B.PID FROM Batting B JOIN ODIMatch M ON B.MatchID = M.MatchID
WHERE M.Team1 = 'Sri Lanka' OR M.Team2 = 'Sri Lanka' GROUP BY B.PID HAVING
AVG(B.NRuns) = (SELECT MAX(AvgScore) FROM (SELECT AVG(B2.NRuns) AS
AvgScore FROM Batting B2 JOIN ODIMatch M2 ON B2.MatchID = M2.MatchID
WHERE M2.Team1 = 'Sri Lanka' OR M2.Team2 = 'Sri Lanka' GROUP BY B2.PID) AS
Averages);
```

```
MySQL localhost:3306 ssl mahek01 SQL > SELECT B.PID FROM Batting B JOIN ODIMatch M ON B.MatchID = M.MatchID WHERE M.Team1 = 'Sri Lanka' OR M.Tea
m2 = 'Sri Lanka' GROUP BY B.PID HAVING AVG(B.NRuns) = (SELECT MAX(AvgScore) FROM (SELECT AVG(B2.NRuns) AS AvgScore FROM Batting B2 JOIN ODIMatch M2
ON B2.MatchID = M2.MatchID WHERE M2.Team1 = 'Sri Lanka' OR M2.Team2 = 'Sri Lanka' GROUP BY B2.PID) AS Averages);
+-----+
| PID |
+-----+
| 25001 |
+-----+
1 row in set (0.0033 sec)
```